

Data Flow Diagram

1. Introduction

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data moves through a system. It identifies the flow of information between users, processes, data storage, and system outputs. In the OptiCrop project, the DFD demonstrates how soil and environmental parameters are processed to generate crop recommendations using a machine learning model.

2. Objective

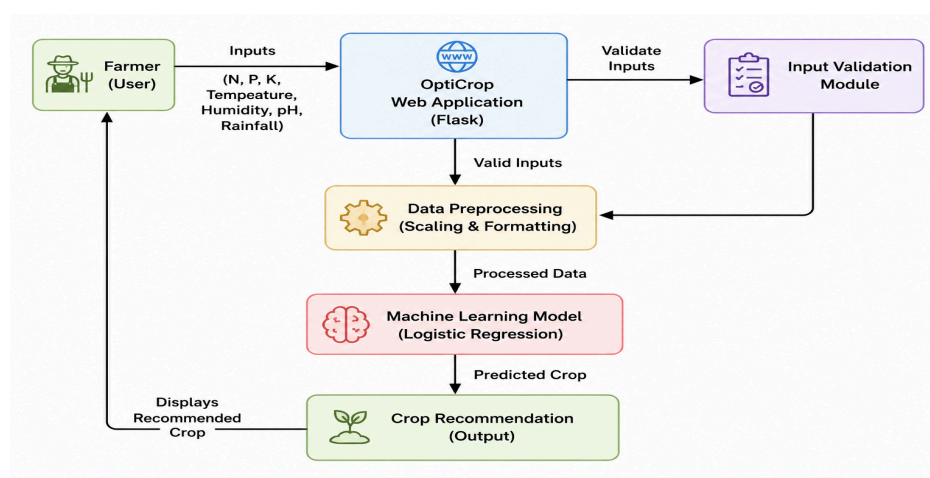
The objective of the Data Flow Diagram is to illustrate the movement of data within the OptiCrop system and explain how user inputs are transformed into meaningful crop recommendations.

3. System Components

The OptiCrop system consists of the following major components:

- User (Farmer)
- Flask Web Application
- Input Validation Module
- Data Preprocessing Module
- Machine Learning Prediction Model
- Crop Recommendation Output

These components work together to process user inputs and generate accurate



recommendations efficiently.

4. Data Flow Description

The overall workflow of the system consists of the following steps:

1. The user opens the OptiCrop web application.
2. Soil and environmental parameters are entered through the prediction form.
3. The application validates the entered values.
4. The input data is standardized using the trained scaler.
5. The processed data is passed to the Logistic Regression model.
6. The predicted crop label is converted into its original crop name.
7. The recommended crop is displayed to the user.

This workflow corresponds to the implemented Flask application, where user inputs are transformed, passed through the trained model, and returned as a crop recommendation.

5. Input Data

The system receives the following input parameters:

Parameter	Description
Nitrogen (N)	Nitrogen content in soil
Phosphorus (P)	Phosphorus content in soil
Potassium (K)	Potassium content in soil
Temperature	Atmospheric temperature (°C)
Humidity	Relative humidity (%)
pH	Soil pH value

Rainfall Annual rainfall (mm)

6. Processing

After receiving the input values, the application performs the following operations:

- Validates the user inputs.
- Organizes the data into the required format.
- Applies feature scaling using the trained StandardScaler.
- Uses the Logistic Regression model to predict the most suitable crop.
- Converts the predicted numerical label into the corresponding crop name.

7. Output

The final output generated by the system is:

- Recommended crop name.
- User-friendly prediction result displayed through the web application.

The recommendation helps users select the most suitable crop according to the given soil and environmental conditions.

8. Advantages of the DFD

The Data Flow Diagram provides several benefits:

- Clearly represents the system workflow.
- Improves understanding of data movement.
- Simplifies system analysis and design.
- Assists in identifying processing stages.
- Supports future maintenance and system enhancement.

9. Conclusion

The Data Flow Diagram provides a clear understanding of how data is processed within the OptiCrop system. It illustrates the transformation of agricultural inputs into meaningful crop recommendations through data preprocessing and machine learning. This structured flow ensures efficient processing, accurate predictions, and an intuitive user experience.